

CSS 343: Data Structures, Algorithms, and Discrete Mathematics

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Overview

- **Instructor:** Eric Reel ericre@uw.edu
- **Office hours:** Tuesday 1:00-2:00pm and Thursday 1:00-2:00pm in UW1-320 or the UW1 third floor lobby.
- **Class Sessions:** 11:00am-1:00pm T/TH in UW1-041
- **Prerequisite:** CSS 301, which may be taken concurrently; and a minimum grade of 2.0 in CSS 342.

Course Catalog Description: Develops competencies associated with problem-solving, algorithms, and computational models. Explores algorithm development and analysis; abstract data types including trees, priority queues, heaps, graphs, and hash tables; use of object-oriented design/programming and design patterns; regular expressions; and language modeling.

Learning Objectives

- An understanding of trees, balanced trees, heaps, hash tables and their uses

- An understanding of the graph data structure and associated algorithms
- Ability to design and implement a complex object-oriented problem
- An understanding of the formal notation for a programming language

Goals

Refining and extending the concepts and skills introduced in CSS 342, students develop competencies associated with problem-solving, design, testing, and programming techniques. Topics include ADTs, data structures, related algorithms, and object-oriented design & programming. Formal automata theory as it applies to programming languages is introduced. Good software engineering and algorithm analysis techniques are used throughout.

As with most technical courses, besides ability and motivation, it takes time to learn and master the subject. No one succeeds without practice! Expect to spend an average of 15 hours a week outside of class time for this course; some of you may spend more time, some less time.

Textbooks

- [Carrano] Frank M. Carrano and Timothy M Henry, Data Abstraction and Problem Solving with C++, Seventh edition, Addison-Wesley, 2017.
- [Cusack] Charles A. Cusack and David A. Santos, [Active Introduction to Discrete Mathematics and Algorithms](#), Version 2.6.3, March 30, 2018.
- [Chen] William Chen, [Discrete Mathematics](#)

- [Rosen] Kenneth H. Rosen, Discrete Mathematics and Its Applications, Seventh Edition, , McGraw Hill, 2011.

Grading

- Exercises: 15%
- Projects: 35%
- Exam-1: 25%
- Exam-2: 25%

Required Course Work

There will be three categories of required course work:

- **Exercises (15%)**

Short assignments and activities some to be performed in-class and some outside of class. Exercises will be graded as Complete/Incomplete. If you are making a good faith effort and submitting the exercise by the due date, you will receive a grade of Complete. Most students who continue to be engaged in the class have an exercise grade of 90-100%. The exercises help prepare you for the exams. Your 2 lowest exercise grades will automatically be dropped in Canvas.

If you are not able to attend class for any reason (personal, professional, COVID quarantine, mental health day, etc), you can email me **before the start of class time** to be exempt from that day's in-class exercises.

- **Projects (35%)**

We will have roughly 5 programming assignments. The programming assignments are sufficiently complex that you should not expect to complete them in one sitting. Start designing and thinking about the project as soon as it is assigned. Ask clarifying questions in class and on discord as necessary. Do not wait until the last day to start your project. Projects have very specific completion and submission requirements. You will need to give yourself enough time to package your project, check that you have completed all the components before submitting. You can submit your project multiple times on Canvas before the due date. I only look at the final submission. Most students have a project grade of 80-100%. The most common reason for low project grade is not starting the project on time. For one and only one project, you can have a 48-hour extension. To indicate you are using the extension, submit a brief explanation to the ["Using Extension"](#) project.

- **Exams (25% + 25%)**

We will have two exams that will be held during the regular class session. The exams are closed-book, closed-notes exam, except for a single cheat sheet (single sheet of letter-sized, double-sided notes created by you). In an exam, you have to demonstrate your knowledge in limited time without any additional resources. The average for exams is usually 70-75%.

Late assignment will only be accepted when it is due to exceptional circumstances such as sickness, bereavement and official university business. I will not make exceptions for work, other classes, personal obligations, etc.

A scale of 90s (3.5■4.0), 80s (2.5■3.4), 70s (1.5■2.4), 60s (0.5■1.4) is a rough guide. A student who achieves 75% will receive a minimum of 2.0, 85% will receive a minimum of 3.0 and 95% and above will receive a 4.0 grade. Scores in between will be interpolated.

<https://canvas.uw.edu/courses/1817334/pages/grade-interpolator>

Inclusion

All students are welcome in CSS 343 and are entitled to be treated respectfully by both classmates and the course staff. I strive to create a challenging but inclusive environment that is conducive to learning for all students. If at any time you feel that you are not experiencing an inclusive environment, or you are made to feel uncomfortable, disrespected, or excluded, please report the incident so that we may address the issue and maintain a supportive and inclusive learning environment. You may contact me or the [CSS Advisors](#) to express your concerns.

Getting Help

There are multiple ways of getting help:

- **QSC**

[Quantitative Skills Center](#) has dedicated hours with 342 tutors that can help. Here is a [video about QSC services](#) and the [tutoring hours](#).

- **Office Hours**

If you need extra time after office hours and cannot get your issues resolved over discord, you can setup a meeting over zoom by sending me email (pisan@uw.edu). I will not debug your code, but I will answer questions and try to help you figure out the problem. I will always ask what you have already done so far to solve the problem, so come prepared.

If you have a personal matter, please see me in office hours.

- **Email**

Please feel free to email me if you have questions or need assistance with an issue. I normally check my email once or twice a day and will do my best to give you a prompt response.

Extenuating Circumstances

I recognize that our students come from varied backgrounds and can have widely-varying circumstances. If you have any unforeseen circumstances that arise during the course, please do not hesitate to contact me to discuss your situation. **The sooner I am made aware, the more easily I can provide accommodations.** While some amount of “productive struggle” is healthy for learning, you should ask me for help if you have been stuck on an issue for a very long time.

Life happens! While our focus is providing an excellent educational environment, our course does not exist in a vacuum. My ultimate goal is to provide you with the ability to be successful, and I encourage you to work with me to make that happen. Please don't suffer in silence.

Academic Integrity

Most exercises and projects in this course will be completed individually. You are here to learn how to code, and that means you actually need to understand how to complete

the exercises on your own. I encourage you to discuss problems with one another or to look for sample code, techniques and reference material on the internet. However, if you are copy/pasting code, you aren't going to be learning anything. Plagiarism will result in an assignment score of zero and a misconduct letter in your student record. Use of resources not explicitly allowed on a test will result in an assignment score of zero and a misconduct letter in your student record. Please be very careful to adhere to the student code of conduct:

<http://www.washington.edu/cssc/for-students/student-code-of-conduct/>. Any code or text that you copy from an external source must have a comment (or reference) indicating its origins. It is also not acceptable for your code or solutions to be publicly accessible on the web (for example, a public GitHub repository).

Access and Accommodations

Your experience in this class is important to me. If you have already established accommodations with Disability Resources for Students (DRS), please communicate your approved accommodations to me at your earliest convenience so we can discuss your needs in this course.

If you have not yet established services through DRS, but have a temporary health condition or permanent disability that requires accommodations (conditions include but not limited to; mental health, attention-related, learning, vision, hearing, physical or health impacts), you are welcome to contact DRS at 425.352.5307 or rosal@uw.edu.

DRS offers resources and coordinates reasonable accommodations for students with disabilities and/or temporary health conditions. Reasonable accommodations are established through an interactive process between you, your instructor(s) and DRS. It is the policy and practice of the University of Washington to create inclusive and accessible learning environments consistent with federal and state law.

For Our Veterans

Welcome! We at UW Bothell understand that the transition into civilian life can be challenging for our veteran students and we have many resources for any who may want to reach out for guidance or assistance. This includes our Vet Corp Navigator through the WDVA and our Student Veterans Association (SVA). Please contact Veteran Services at 425.352.5307 or rosal@uw.edu. For those of you needing more URGENT support, please call The Suicide Prevention Hotline 1.800.273.8258 or connect with the UWB CARE Team <https://www.uwb.edu/studentaffairs/care-team>.

Religious Accommodations

Washington state law requires that UW develop a policy for accommodation of student absences or significant hardship due to reasons of faith or conscience, or for organized religious activities. The UW's policy, including more information about how to request an accommodation, is available at [Religious Accommodations Policy](#). Accommodations must be requested within the first two weeks of this course using the [Religious Accommodations Request form](#).

Common Course Policies for the School of STEM

See https://bit.ly/STEMClassroom_Policies for additional information on Academic Integrity, Access and Accommodations, Classroom Emergency Preparedness, For Our Veterans, Grade of Incomplete, Inclement Weather, Parenting Resources, Respect for Diversity. Student Support Services, Wondering How to Address Faculty? etc. (P.S. I prefer to be addressed as 'Professor Pisan')

Internships

Internships: One of the strengths of the computing industry is a culture that values internships. As a result, many industry leaders have well-developed internship programs for students to build their expertise and pathways to opportunities. Internships are offered throughout the year, some are intended for students just starting out (or just exploring) the technology industry, and many for the summer between your junior and senior year. **We strongly encourage you to participate in internships and attend campus Career Fairs and employer events to connect with industry partners and learn more about opportunities, hiring timelines and application processes.**

The common technical recruiting cycle is to hire in the fall for the next summer's internships and jobs. For some internships, this means the technical interview occurs 6-8 months before the actual internship. Thus, it is important to convey (in your application and interview) what your current knowledge base is at the time of the interview, and what it will be at the time of the internship. Because both of the Data Structures and Algorithms courses prepare you best for technical interviews, we recommend you do the following to be successful in the fall recruiting process:

- Check what the expectations are for the internship you are applying for. This includes both expectations at the time of the internship as well as at the time of the interview. The majority of internships are meant for juniors, but some are meant for graduating seniors.
- **Clearly communicate your qualifications**, being careful to accurately list your preparation and qualifications. It is important that you are able to back up the claims you make about your experience and skills from your application materials in behavioral and technical interviews. Misrepresentation of these reflects poorly on candidates and their future opportunities.
- We encourage you to take CSS 342 the fall quarter before the summer internship to prepare you better for the technical interviews.

- On your resume, list each course and the date you will be taking them. Thus, a student starting in Fall 2022 would list, for example: Data Structures, Algorithms, and Discrete Mathematics I (completing Fall 2022), Data Structures, Algorithms, and Discrete Mathematics II (completing Winter 2023).
- When you are invited for a technical interview, let the recruiter know where you are in the course series.
- To fill-in your content knowledge, prepare and practice for your technical interviews by utilizing the resources provided by Career Services at <https://www.uwb.edu/career-services/interviews/technical-interviews>.

There are many students who use internships for their Capstone course. However, this is **not** the only time when a student can do an internship. Rather, we encourage students to participate in **more than one experiential learning opportunity**, and internships can be done independent of receiving credit. If you decide to do a non-Capstone internship for credit, you can receive up to 5 cr of CSS 397, which will count as a 200-level CSS elective.

Finally, as a reminder, if you are considering an internship for your Capstone and you are a:

- CSSE major: You must first complete the Capstone prerequisites in order to use your internship for CSS 497. Prerequisites are completion of all core courses and 2 CSS electives. Most CSSE students will **not** have met the capstone prerequisites by their first summer. As mentioned earlier, these prerequisites are not a requirement for doing a non-Capstone internship (CSS 397) for credit.
- Applied Computing major: You cannot use an internship for your Capstone. You can, however, use an internship to earn up to 10 cr of CSS 495 which will count towards your 400 level CSS electives.

Your faculty, advisors, and Career Services are all great resources to help you be successful in your career pursuits.

- Make an appointment with your FYPP Academic Advisor (uwbadvis@uw.edu) or a CSS Academic Advisor (cssadv@uw.edu).

Make an appointment with Career Services
at <https://www.uwb.edu/career-services/appointments>.